

Calibration model

$$Y_n = a_0 + a_1 X_n + E_n \quad (10.1)$$

Extrapolated concentration

$$Z^* = \frac{a_0}{a_1} \quad (10.2)$$

Variance of the extrapolated concentration

$$s^2(Z^*) = \left(\frac{s_E}{a_1} \right)^2 \times \left[\frac{1}{N} + \frac{\bar{Y}^2}{a_1^2 \sum_n (X_n - \bar{X})^2} \right] \quad (10.3)$$

Residual standard deviation of the regression

$$s_E = \sqrt{\frac{\sum_n (Y_n - \hat{Y}_n)^2}{N - 2}} \quad (10.4)$$

The calculation is repeated at each wavelength, and finally, three extrapolated measurement values, denoted Z_j^* with $1 \leq j \leq J$ are collected. Each corresponds to three measurements: one without spikes and two with different spikes. These measurements will be assumed to be replicates under repeatability conditions, as they cannot be considered as obtained under intermediate precision conditions since the measuring techniques are not identical, since different wavelengths are used.

The regression line coefficients are estimated by OLS regression. The three extrapolated concentrations must be multiplied by the dilution factor of 1: 2500 to recover the expected contents in the tablet. For each extrapolated concentration, it is also possible to calculate a variance using the formula (10.3) proposed by [5]. The number of associated degrees of freedom is then ridiculously small and equal to 1. The variance is used as an estimate of the standard variance of an extrapolated result $s^2(Z^*) = u^2(Z^*)$.

The implementation of the formulas in Excel® is simple when using the built-in matrix function `LINEST` because it directly provides the residual standard deviation s_E as explained in Section 2.3.1 and shown on the Resource Q worksheet. To use `LINEST` in its matrix form, please refer to the Excel® user's manual.

For the illustrated application of Resource Q, the calculation for the SAM is done for a single wavelength at 200 nm. The results returned by `LINEST` are surrounded

Table 10.1 TDF measurements in a drug declared at 245 mg/tablet, obtained by HPSAM at different wavelengths ($1 \leq j \leq 3$).

Wavelengths (nm)	Extrapolated content (mg/ml)	Variance	Level found (mg/tablet)	Recovery rate (%)
200	0.0977	1.111×10^{-5}	244.1	99.6
210	0.0973	5.567×10^{-5}	243.2	99.3
240	0.0967	2.303×10^{-6}	241.8	98.7
Average $\bar{Z}$	0.0972			99.2

by a light line, and the residual standard deviation $s_E = 0.00777$ by a thick border in cell C13. The most complicated formula is the Intermediate quantity in cell B22. It corresponds to the quantity between square brackets in Eq. (10.3).

Resource Q Calculation of the SAM extrapolated concentration (Excel).

	A	B	C	D
1	Resource Q: Calculation of the extrapolated concentration by SAM			
2	Tenofovir disoproxil (TDF)			
3	Dilution factor	2500		
4	Nominal value (mg/tablet)	245		
5			Optical density (OD)	
6		Spike in mg/ml	DO at 200 nm	
7		0.000	0.50	
8		0.049	0.76	
9		0.098	1.00	
10				
11	SAM calibration line	Slope a1	Intercept a0	Formulas
12		5.115	0.499	{=LINEST(C7:C9:B7:B9;1;1)}
13		0.1121	0.0071	{=LINEST(C7:C9:B7:B9;1;1)}
14		0.9995	0.00777	{=LINEST(C7:C9:B7:B9;1;1)}
15		2080.6	1.000	{=LINEST(C7:C9:B7:B9;1;1)}
16		1.26E-01	6.04E-05	{=LINEST(C7:C9:B7:B9;1;1)}
17				
18	Extrapolated concentration (Z*)	0.0977	=C12/B12	
19	Average Z	0.7501	=AVERAGE(C7:C9)	
20	Number of spikes (N)	3	=COUNTA(B7:B9)	
21	se/a1	0.001519	=C14/B12	
22	1/N	0.3333	=1/B20	
23	Intermediate quantity	4.8123	=B22+((B19*B19)/(B12*B12*DEVSQ(B7:B9)))	
24	s(Z*)	3.3327E-03	=B21*SQRT(B23)	
25	s ² (Z*)	1.1107E-05	=B24^2	
26	Recovered conc. (mg/tablet)	244.1	=B18*B3	
27	Recovery yield	99.6%	=B26/B4	
28				

The same worksheet is copied and updated for each wavelength. The final results for the three wavelengths are gathered in Table 10.1. Recovery rates are obtained by accounting for the dilution factor applied to the extrapolated contents compared to the content announced by the manufacturer.

Finally, $J = 3$ replicate results are combined, each with an estimated standard variance, i.e. squared standard uncertainty. The addition of covariances is probably necessary, but it is impossible to calculate them, given the small number of degrees of freedom. Since each measurement value is considered one replicate, the analytical result can be expressed as an average. By making this approximation, it is also possible to estimate the repeatability variance by taking the average of the variances of the replicates (formula 10.5).

The resulting situation is comparable to the one described in Section 8.4.3 about calculating the MU of a result obtained by averaging replicates achieved under repeatability conditions. This gives the following formulas:

Mean extrapolated concentration

$$\bar{Z} = \frac{\sum_{j=1}^J Z_j^*}{J} \quad (10.5)$$

Repeatability variance of the mean

$$s_r^2 = \frac{\sum_{j=1}^J s^2(Z_j^*)}{J} \quad (10.6)$$

Standard variance of the mean content

$$u^2(\bar{Z}) = \frac{s_r^2}{J} = \frac{\sum_{j=1}^J s^2(Z_j^*)}{J} / J \quad (10.7)$$

Numerical application to the results for the TDF analysis gives:

Parameters	Extrapolated value	Predicted concentration	Recovery yield
Unit	mg/ml	mg/tablet	%
J	3		
$\bar{Z}$	0.0972	243.0	97.22
s_r^2	2.303×10^{-5}		
$u^2(\bar{Z})$	7.676×10^{-6}		
$U(\bar{Z})$	0.00554	13.9	
$UR\%(\bar{Z})$	5.7%	5.7%	
Coverage interval	0.0917	229.2	91.68
	0.1028	256.9	102.76

Assuming that the variance in formula (10.7) is the standard variance may be abusive, insofar as it was obtained under repeatability conditions. But the small number of degrees of freedom is a limitation that impairs the quality of this estimate. Nevertheless, it represents a first estimate of the MU that is easily computed. The relative uncertainty thus obtained, close to 6%, is relatively high if one considers the $\pm 5\%$ acceptance criterion classically adopted in pharmacy for active principles but it can be considered operative in the present context.

Figure 10.2 shows a similar study performed on a drug containing FTC. In this case, the standard additions are slightly different (0.040 and 0.080 mg/ml) to fit the expected nominal content, declared at 200 mg/tablet.

The dilution factor is the same. The results obtained from these data are as follows:

Symbol	Extrapolated FTC content (mg/ml)	Level found (mg/tablet)	Recovery yield (%)
J	3		
$\bar{Z}$	0.0799	199.9	99.94
s_r^2	3.612×10^{-5}		
$u^2(\bar{Z})$	1.204×10^{-5}		
$U(\bar{Z})$	0.00694		
$UR\%(\bar{Z})$	8.7%		
Coverage interval	0.0730	182.5	91.26
	0.0869	217.2	108.61

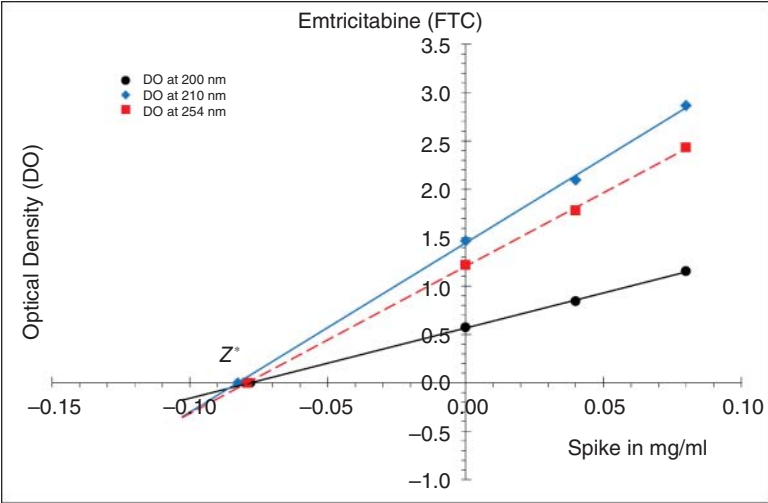


Figure 10.2 Determination of FTC, by standard additions to a pharmaceutical product announced at 200 mg/tablet.

In this example for FTC, the analytical method achieves an inferior performance since the estimated relative uncertainty is around 9%. The proposed H-point procedure and calculation method were applied to six imported pharmaceutical preparations containing TDF and FTC. Estimated relative uncertainties range from 4 to 19%. In each case, the inverse-predicted mean content is calculated from 9 absorbance measurements, as well as its coverage interval.

Figure 10.3 summarizes these results. On the same graph, the $\pm 5\%$ acceptance limits classically used in the pharmaceutical field for formulations are plotted in

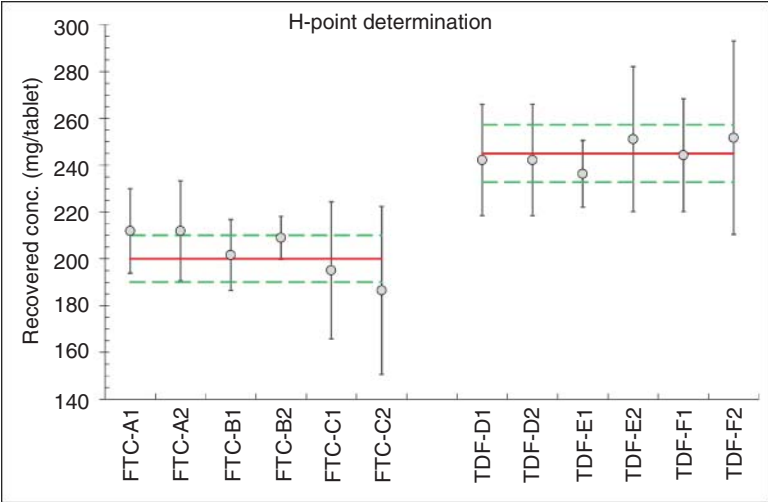


Figure 10.3 Average levels found (mg/tablet) with the coverage intervals for six medicines noted from A to F. Expected nominal content is represented by broad red lines surrounded by the acceptance interval of $\pm 5\%$ as green dotted lines.

green dotted lines. It appears that for several controls, such as FTC-C2, the situation is quite degraded.

10.1.2 SAM with Replication

In Figure 10.3, the full replication number of the analysis is indicated by the last digit of the label. Finally, each of the six samples, three containing FTC and three with TDF, were independently duplicated. Thus, the experimental design includes several series of measurements. It is then possible to extract a between-series variance and calculate the uncertainty of the inverse-predicted concentration in closer accordance with the GUM recommendations.

As stated in Section 8.4.3, it is not an optimal experimental design since there are only two series of three replicates, while it is recommended to make three series of two replicates. The notations used for the experimental design are as follows:

I	Number of series $I = 2$
J	Number of replicates per series $J = 3$
Z_{ij}^*	Extrapolated content (at a wavelength q) for a series I
$\overline{Z}_i^*$	The average content of series I (Eq. 10.5)
$s^2(Z_{ij}^*)$	The variance of an extrapolated content (Eq. 10.6)
T	Inverse-predicted concentration in a tablet, $T = \overline{\overline{Z}} \times \frac{1}{\text{Dilution Factor}}$

This occurs to the following formulas:

Grand mean

$$\overline{\overline{Z}} = \frac{\sum_{i=1}^I \overline{Z}_i^*}{I} \quad (10.8)$$

Repeatability variance

$$s_r^2 = \frac{\sum_{i=1}^I \sum_{j=1}^J s^2(Z)_{ij}^*}{IJ} \quad (10.9)$$

Between-series variance

$$s_B^2 = \frac{\sum_{i=1}^I (\overline{Z}_i^* - \overline{\overline{Z}})^2}{I - 1} \quad (10.10)$$

Standard variance of the grand mean

$$u^2(\overline{\overline{Z}}) = \frac{s_B^2}{I} + \frac{s_r^2}{IJ} \quad (10.11)$$

Numerical application of these formulas to six batches of drugs labeled A to F from two manufacturers gave the following results, grouped in Table 10.2.

Figure 10.4 illustrates these results. In general, the inverse-predicted content is obtained with an almost acceptable uncertainty if one considers the classic acceptance range of the pharmaceutical field for this determination.